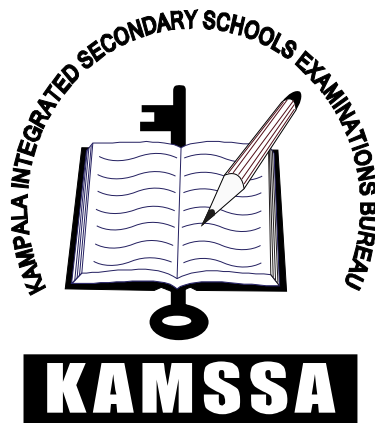




SET ONE
KAMSSA PRE REGISTRATION EXAMINATIONS 2025
Uganda Certificate of Education
PHYSICS
Paper 1
Time: 2 Hours 30 minutes

INSTRUCTION

- Attempt all items

ITEM 1

James and John are friends who were having a game while standing a distance from each other on a road. James called John using the mouth but John could not hear easily. Later they developed two open tins with a long string connecting the tins. James put the tin on his mouth calling John who put the tin on his ear. John heard the sound very fast than before. Looking beyond John along the road, James saw a pool of water which made them run there, and unfortunately, they never saw the pool on the road. They wondered where the pool had gone.

Hint;

- Speed of sound in air is 340ms^{-1} .
- James and John were 40 m apart.
- It took 0.0125seconds for John to hear sound through the string.

Task

As a learner of Physics, help John and James understand;

- a) There was delay for sound to be heard while using the mouth.
- b) The sound was heard faster with the tins and string.
- c) There was variation in the speeds of sound.
- d) There pool of water was seen and disappeared.

ITEM 2

Astronauts enjoy exploring space. However, they say that there are some planets they can land on and others not. They have visited Mars but claim it is not fit for life like earth. Using telescopes, they have viewed stars that twinkle with different colours and produce heat. One of the astronauts said that stars are "born, grow and die, " which confused the rest.

Task

- As a learner of physics, help one understand why,
- it is safe to land on some planets while others are not
 - Mars is not suitable for life like the earth.
 - stars twinkle with different colours.
 - the statement " stars are born, grow and die" is valid.

ITEM 3

Mary is a new mother with a baby boy. She has to bathe the baby water ranging from 35°C to 40°C so as not to burn the baby's skin. While sitting in a vehicle back home, the engine temperature rose and the driver requested for 5 liters of cold water to pour in a black container fitted with some metal fins. At home, Mary boiled 2 liters of water up to 80°C and mixed with 4 liters of cold water at 20°C in a metallic can weighing 3kg.

Hint;

- Density of water = 1000kgm^{-3}
- Specific heat capacity of the can = $400\text{JKg}^{-1}\text{K}^{-1}$
- Specific heat capacity of water = $4200\text{JKg}^{-1}\text{K}^{-1}$

Task

- As a learner of Physics,
- help Mary know if the water she cooked was suitable for bathing the baby.
 - why did the driver pour water in the black container but not any other liquid.
 - why was the container black with metallic fins?

ITEM 4

A given nursery school has intentions of putting up swings for children whose mechanical energy should not exceed 375J. The swing is to be put at the vertical height of 0.6m at equilibrium position. It is known that as children swing, heat is produced at rope contact areas which might accumulate and become harmful to children in case it exceeds 46°C and the rope may burn.

Hint;

- All mechanical energy of the system changes to heat energy.
- Average temperature at contact points is 20°C
- Heat capacity of ropes materials is 20JK^{-1}
- Average mass of child is 25kg
- Acceleration due to gravity = 10ms^{-1}

Task

- As a learner of physics, help one
- know the maximum height a child must go from equilibrium.
 - the maximum speed of a child past equilibrium position
 - know whether the rope is appropriate for use
 - know the source of heat and how it can be minimized.

ITEM 5

A boy walked to a shop to buy flour and reached thirsty. He decided to first buy a bottle of cold water and removed the polythene seal on the lid. He observed the seal attracted to the fur on his fingers something which surprised him. He later bought the flour and saw some metallic specs in the flour which worried him. The attendant advised him that using a long wire, a pair of dry cells, and a long iron rod, the flour would be sorted but the boy didn't know how to carry out the process.

Task

As a learner of physics, help the boy understand

- a) how the seal was able to be attracted to his fingers.
- b) the electric field interaction between the boy and the seal
- c) How he could sort the flour
- d) how the effect is helpful in sorting the flour would be increased.

END